

Perspective: Performance Loss Rate in Photovoltaic Systems

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Photovoltaic systems may underperform expectations for several reasons, including inaccurate initial estimates, suboptimal operations and maintenance, or component degradation. Accurate assessment of these loss factors aids in addressing root causes of underperformance and in realizing accurate expectations and models. The performance loss rate (PLR) is a commonly cited high-level metric for the change in system output over time, but there is no precise, standard definition. Herein, an annualized definition of PLR that is inclusive of all loss factors and that can capture nonlinear changes to performance over time is proposed. The importance of distinguishing between recoverable and nonrecoverable losses which underly PLR is highlighted.

1. Introduction

PV systems remain in service for decades, making it important to plan for performance changes. There are a wide variety of factors that affect performance causing it to change over time. Quantifying the impacts of these losses is critical to reducing the financial risks of PV systems.

In this perspective, we present a framework for quantifying these loss factors and the complexities associated with their estimation from field time-series. We begin by proposing a precise definition of the term performance loss rate (PLR) and related concepts. PLR is often cited as a key performance

indicator of PV system health that quantifies changes in output over time, but we are aware of no standardized definition for PLR. Further, many studies conflate PLR and component degradation (i.e., module, DC array, etc.), which we argue are distinct but related concepts.

PLR is a high-level metric capturing all the losses a system experiences, but it does not provide actionable information on its own. Only by evaluating the individual losses that underly PLR can actionable information about correctable issues or the performance of different technologies be determined.^[1] Thus, PLR provides a starting point for disentangling these underlying losses. It is important for researchers and analysts to be precise when describing the individual loss factors underlying the high-level PLR metric. These values are quantified by a wide variety of analysis algorithms or measurement campaigns, but comparing and using results from different methodologies is challenging. Furthermore, loss factors often confound one another, and it can be challenging to disentangle them. Tools such as the TRUST-PV Risk Matrix are helpful in that they provide a common and specific framework for referencing losses and allow observations to be coded for subsequent analysis.^[2] Further, we emphasize that losses can also be broadly categorized as either recoverable or nonrecoverable. This distinction is particularly important because the two types of losses have different implications. Recoverable losses (e.g., soiling, snow, shading due to vegetation, etc.) can potentially be addressed through a change to operations and maintenance (O&M) activities, while nonrecoverable losses (e.g., material degradation) can be quantified to assess risks associated with different technologies in different environments. The distinction is also important because recoverable and nonrecoverable losses can influence the high-level PLR metric in different ways.

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Within this context, we argue that practitioners should be precise about which losses are in and out of scope for their analyses and studies. We present examples of this for three published analyses.

2. Performance Loss Rate

We begin with precise definitions of PLR and supporting concepts. Existing descriptions of PLR in the literature describe PLR as quantifying all losses, both recoverable and nonrecoverable.^[3,4] Examples of included loss factors are component degradation, snow loss, soiling, and outages. However, there is no formalized and accepted definition of PLR. In this section, we propose a precise annualized definition.

Two common metrics for assessing the performance of PV systems are the performance ratio (PR) and performance index (PI). These quantities are formalized in IEC 61 724. Briefly, PR is the ratio of a system's energy yield to a reference yield calculated from the total insolation received. PR can also be calculated with a correction for module temperature to help mitigate seasonality. Module temperature can be accounted for with either direct measurements or a model accounting for irradiance, ambient temperature, and wind speed. PI is the ratio of the measured system output to the expected output from a model of the system given the measured irradiance and weather. PI can reflect more system details than PR. For example, systems are commonly designed with a DC/AC ratio above 1, meaning that the DC capacity of the system exceeds the inverter capacity. This results in "clipping" during times of high irradiance when the modules are not operated at their maximum power point. The model for expected output in PI can account for clipping, but PR calculations do not include any information about DC/AC ratio. Thus clipping would reduce PR, but not PI. In this way, PI can more closely track changes in the system while accounting for its design; the same is true for other system-specific details.

PI can be calculated with power as the power performance index (PPI) or energy as the energy performance index (EPI). Here we focus on a special case of EPI, which we term the simple energy performance index (SEPI). SEPI relies on a model that utilizes exclusively fixed loss factors that do not change over time. That is, SEPI is calculated from a model that has constant losses for things like soiling, snow, mismatch, and availability which do not change from year to year or based on weather observations. PVWatts is an example of such a static model.^[5] As an example, soiling may be included in a static model as an annual or monthly derate, but could also be included as a sawtooth profile (per the Kimber model^[6]), as long as the same soiling profile is used every year. Further, the model used to calculate SEPI does not include degradation. The model used for SEPI does account for actual observed irradiance and can account for temperature.

We define PLR as the change in annualized SEPI relative to the first year. Specifically, the PLR of year i is given by

$$PLR_i = \frac{SEPI_i - SEPI_{i-1}}{SEPI_1} \quad (1)$$

where i starts at year 2 because this formulation of PLR is undefined until the conclusion of the second year.

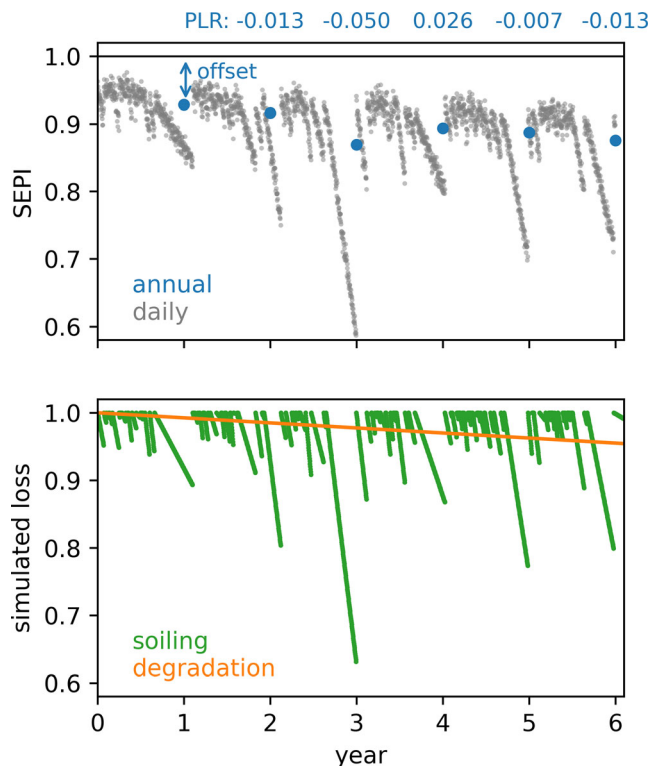


Figure 1. Example SEPI and PLR for synthetic PV data for a system affected with soiling and linear DC degradation. The upper panel shows annualized and daily SEPI. The SEPI of year 1 does not have a PLR because PLR is defined relative to the prior year's SEPI; the initial underperformance relative to a SEPI of 1 is an offset relative to the performance model and does not impact PLR. The PLR for subsequent years is shown above the plot (e.g., a value of -0.013 represents a 1.3% loss from the previous year). The lower panel shows the daily losses from soiling and DC degradation used to synthesize the dataset.

An example of SEPI and the associated PLR series is shown in **Figure 1** with synthesized PV power production. PV energy output was simulated for a single axis-tracked system, with a DC/AC ratio of 0.9 in Golden, Colorado, USA, based on historical satellite irradiance data. To model losses, the idealized model was multiplied by signal components representing Gaussian noise, soiling, and DC degradation. It was also reduced by 5% to illustrate an offset between expected and realized output. The signal components were generated following the methods of Skomedal and Deceglie^[7] with a degradation rate of $0.75\% \text{ year}^{-1}$ and randomized soiling with a median soiling rate of $0.2\% \text{ day}^{-1}$, standard deviation of $0.1\% \text{ day}^{-1}$, and 300 cleaning events. Figure 1 illustrates the distinction between the model offset and PLR. The offset is a difference in performance relative to the model, while PLR exclusively quantifies changes and is normalized to the SEPI of the first year, regardless of model offset. The example also shows how PLR can fluctuate from year to year due to changes in loss factors, in this case soiling. For example, we see a rapid PLR in year 3 due to severe soiling losses; this is followed by a positive PLR in the next year which was less affected by soiling.

There are several important features of this formulation of PLR that we wish to highlight. First, this definition of PLR is

inherently annualized. It can take on a different value in each year and is not defined until the end of the second year. Often, degradation and PLR are assumed to be linear. These approaches are compatible with this definition, they just imply a constant PLR each year. However, the definition proposed here also allows for nonlinear changes in performance over time.^[8]

Another important feature arises from the fact that PLR depends on the simple performance index. The PLR captures all changes in performance except those arising from variations in solar resource and module temperature. This is consistent with prior descriptions of PLR.^[3,4] For example, if there is more snow accumulating on panels in one year than in the previous year, this would be reflected in a higher PLR for the snowier year. We also note that some loss factors can be reduced in any given year, resulting in a positive PLR for that year. For example, consider the soiling losses in the third year versus fourth year of the example shown in Figure 1. Another example is that the PI in the first year of operation can be low as the system is brought online, this can lead to both a positive PLR for year 2; this would also affect PLR of subsequent years because year 1 SEPI is the denominator in the definition of PLR.

Though PLR captures all changes to performance, it does not capture offsets relative to the initial model. Such an initial offset is illustrated by the first year's SEPI in Figure 1. In addition, PLR does not capture performance relative to a nonstatic performance model (e.g., one that includes an assumed PV degradation rate). Differences between realized system performance and such models are consequential for stakeholders. However, it is important to be clear that they are distinct from PLR, which exclusively quantifies changes relative to initial performance.

3. Practical Considerations in Calculating PLR

Like PI, PLR is specific to the performance model and data sources used to calculate expected power. The details of the model must be specified when reporting PLR. In many applications, researchers work with limited or imperfect metadata about systems. In these cases, they may choose to implement a very simple energy yield model. Even a very simple model accounting just for irradiance and temperature can be used, but it may yield different PLR results compared to a model that utilized detailed information about the system; the AC/DC ratio, spectral correction, and tracking strategy are examples of details that could be included in a detailed model.^[9] Similarly, it is important to consider different irradiance sources and their errors. Generally, higher quality data will result in a more accurate PLR estimate. For example, use of on-site irradiance is attractive, but if the sensors are not regularly cleaned, it may prevent the proper quantification of soiling losses. In another example, an irradiance tracker on a broken tracker would potentially mask the performance issue of the broken tracker. The details of the data sources are critical to the interpretation of reported PLR. Like the model used for calculating SEPI, they should be specified alongside PLR and well documented in all reporting.

This definition of PLR is simple but estimating it with real-world data is much more complex. It is often not advisable to simply integrate energy and perform the calculation. The definition

as presented here should be taken as a fundamental definition rather than a prescription for calculation. In real-world data sets, outages and erroneous data must be considered and handled. An extensive discussion of these issues and approaches can be found in the literature.^[3,4,10] For example, if a system experiences data outages for a portion of a year plagued by loss from snow-covered modules; the absence of those points would bias the SEPI high and likewise influence comparison to neighboring years. Where possible, it is advisable to carefully correct or impute (and not necessarily remove) problematic data when estimating PLR.

Although data filtering and correction can be useful in estimating PLR, it is important not to filter out real losses that should be counted. Again, PLR is a metric intended to capture changes in time for all loss factors. It is important not to remove or correct out real system losses like full or partial outages, snow, and soiling. Similarly, times of clipping should not be removed because DC/AC ratio is an important system characteristic, and its effect on the relationship between DC degradation and PLR should be reflected. The preceding discussion applies to the basic definition of PLR. In some applications, it may be desirable to estimate and report PLR after removing or correcting for specific loss factors. In these cases, we suggest adopting the nomenclature of (loss)-corrected PLR, for example, soiling-corrected PLR.

4. Factors Underlying PLR

PLR as defined here accounts for changes to all the losses a system may experience. PLR is useful as a high-level metric, but it does not provide any diagnostic actionable insights into how to improve system performance. There are a wide variety of losses which contribute to PLR. Broadly, they can be categorized into two groups: losses that can be recovered naturally or through some type of maintenance and losses that are nonrecoverable. Examples of loss factors categorized in this way are shown in Figure 2. Both types of loss factors contribute to PLR.

Recoverable losses include issues that can resolve themselves or be addressed through maintenance on the system. These losses can be recovered in principle, but in many cases, recovery is impractical, difficult, or impossible, for example, mitigating shade sources on neighboring properties. Curtailment is another loss that is classified as recoverable but is outside the direct control of the system owner. Many of these losses involve attenuation of the solar resource reaching the solar cells. Such factors are soiling, snow, tracker outage or pointing error, and shade. Others involve components themselves, including outages, availability, and inverter performance. An example of a recoverable inverter performance issue is clogging of the air filters resulting in overheating and derated performance. Constant estimates for recoverable losses can generally be captured in the static model underlying SEPI. However, by definition, such a model must account for them in a way that does not change year after year. If operations and maintenance are ineffective, these losses can increase over time and contribute to a rapid PLR. Similarly, environmental changes affecting naturally cleaned soiling or snow can lead to changes in PLR from year to year.

In contrast, degradation of the AC and DC systems is generally categorized as unrecoverable, meaning it is related to a permanent physical change to the system that cannot be recovered

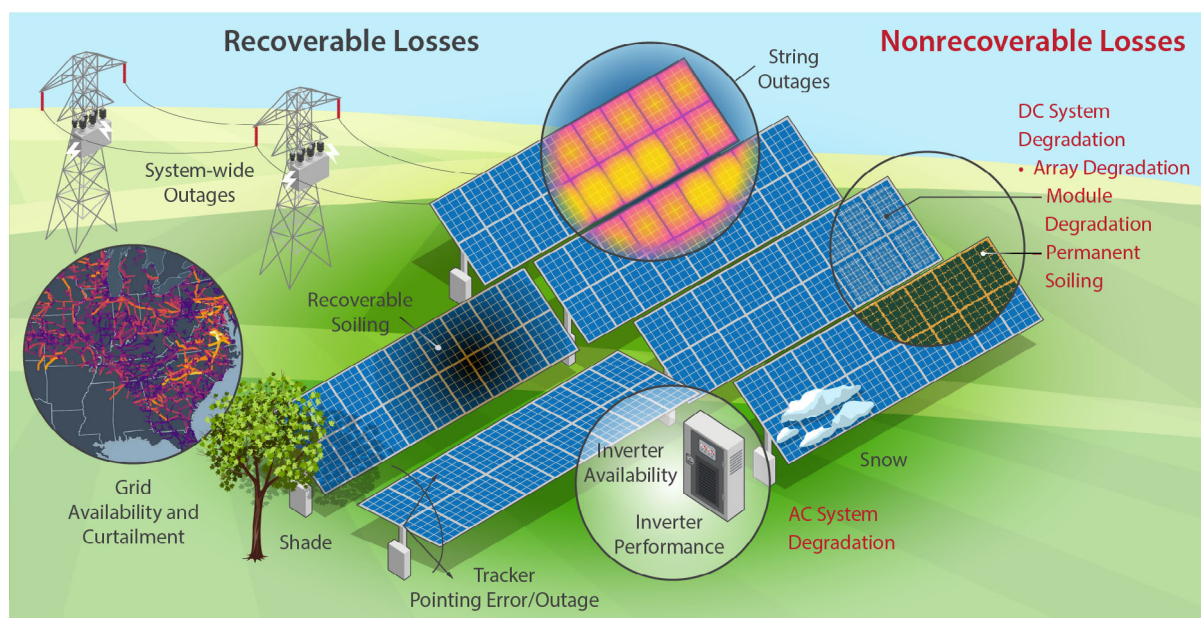


Figure 2. Recoverable (black text) and nonrecoverable (red text) losses. Recoverable losses are those that can be mitigated in principle, although it is not always practical to mitigate them entirely in practice. Both recoverable and nonrecoverable losses contribute to PLR.

without wholesale replacement of components. It is important not to confuse the PLR, a higher-level metric, and degradation, which is one of many factors underlying PLR. For DC degradation in particular, the actual impact on PLR depends on a hierarchy of effects and the specifics of the system. At the lowest level, PV modules degrade with time. However, even among identical modules, degradation proceeds according to a statistical distribution. Over time, electrical mismatch between modules in an array can cause degradation on a subset of modules to have an outsized impact on the overall performance of the array. Multiple arrays may ultimately make up a single system, and the details of the electrical topology dictate how degradation ultimately affects PLR. In particular, the use of string inverters or module-level power electronics affects mismatch and would be expected to change the way in which module degradation ultimately contributes to PLR. Further, the DC/AC ratio of a system affects the relationship between PLR and DC degradation. In a system with a high DC/AC ratio, clipping partially mitigates the impact of module degradation, as the modules can degrade without affecting the peak power output of the system. The effect of clipping on AC output over time is illustrated for a simulated system in **Figure 3** which shows the evolution of annual AC output for a system with annual DC degradation of $-0.75\% \text{ year}^{-1}$, DC/AC ratio of 1.3, simulated using pvlb,^[11] and typical meteorological year data for northeastern Colorado. The degradation does however cause some annual AC loss by reducing output during nonclipped operation and reducing the amount of time spent at peak power output.

We acknowledge a certain vagueness with this distinction between recoverable and nonrecoverable losses, because ultimately everything in a system can be replaced. We suggest the distinction should be made at what is reasonable or practical to address through operations and maintenance including

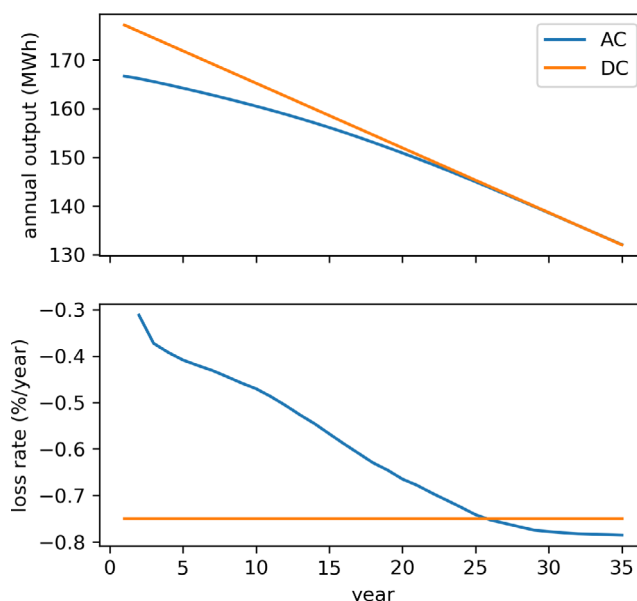


Figure 3. Top: Simulated annual energy output for a PV system with a DC/AC ratio of 1.3 and DC degradation of $0.75\% \text{ year}^{-1}$. Bottom: Realized loss rate on the AC and DC side of the system relative to the first year. Clipping initially mitigates impact of DC degradation because modules can degrade without affecting peak output. Eventually degradation causes peak output to fall below the max AC rating and the AC loss rate levels off near the DC loss rate. Because both loss rates are relative to year 1 and the initial AC capacity is less than the initial DC capacity, the AC loss rate levels are slightly below the DC loss rate.

repairs to system components but short of repowering or redeveloping the system. We also acknowledge that some recoverable losses may not be economical to recover in practice.

Soiling provides a useful example illustrating how loss factors may be convoluted with one another. Soiling is fundamentally a resource attenuation effect; however, sometimes soiling accumulates on modules and is not, or cannot be, removed. In cases where the soiling loss is not recovered due to either natural or O&M cleaning, its effect on performance may be indistinguishable from that of module degradation. Further, if soiling occurs nonuniformly over an array, it can contribute to mismatch losses. It is important to be clear when presenting, for example, soiling losses, exactly what metric is being reported. A time series analysis of PV system production may only be able to estimate the recovered impact of soiling and lump permanent soiling in with degradation. On the other hand, a study in which individual modules are cleaned and tested may capture soiling losses that are never recovered in practice, especially if the system O&M does not include cleaning. These two different quantifications of soiling are both valid, but are not directly comparable. It is critical for researchers and analysis to be clear about the boundaries and assumptions of their approaches to quantifying loss factors. The framework presented here supports this.

5. Examples

There are a wide variety of analyses and methods used to quantify loss over time and the underlying loss factors. It is important to be clear about whether any given analysis is assessing high-level PLR or the underlying factors. Further, the underlying loss factors often confound one another or are challenging to disentangle. It may not be straightforward to compare estimates of nominally the same loss factor if the methodologies are substantially different.

In this section, we describe several analyses with special attention to how they differ and what losses they include. We encourage practitioners to critically examine their own analyses and document a description of which losses or combination of losses are quantified. This provides others with the appropriate context to develop appropriate model inputs based on the results.

Bolinger et al.^[12] used annual net generation data for over 20 GW of PV along with a fixed-effect model to assess fleet-wide changes in performance over time. Since this analysis was based on annual generation relative to a model, it provides an estimate of a PLR (assuming a constant-PLR model). However in the final incarnation of the analysis, the authors explicitly remove the effects of curtailment. In this case, the analysis estimates a curtailment-corrected PLR.

In contrast, other published analyses focus on DC degradation rather than high-level PLR. Two examples are the PV Fleet Data Initiative^[13] and the statistical clear sky approach.^[14]

The PV Fleet Data Initiative^[13] aggregates data from system owners to provide insights into the distribution of degradation rates experienced across the US fleet of PV systems. The initiative utilizes RdTools^[15] to extract degradation rates for each inverter in a system. Briefly, the analysis begins with PV production data at relatively high temporal resolution, typically 15 min. At this resolution, PV production for each data point is normalized to expected energy based on a simple model accounting for module temperature and irradiance. Data is then filtered to remove performance index points below 0.01, irradiance outside the range 200–1200 W m⁻², anomalous temperature readings,

and clipping. Full system outages are also excluded. After filtering, the data is aggregated daily using an irradiance-weighted average. Finally, the long-term degradation trend is estimated using the year-on-year method.

The use of filtering in this way implies that the analysis is primarily targeting an estimate of DC degradation. Outages at the inverter level are explicitly excluded from the analysis; energy lost to such outages is not quantified. Other partial availability issues are not explicitly excluded. If these issues have a trend in time, they will affect the degradation rate estimate. If on the other hand they do not have a long-term trend, they simply contribute to a larger confidence interval.

Another important distinction between a PLR estimate and the Fleet Data Initiative approach is that PLR is inherently weighted by seasonal resource variations. PLR is affected more by losses during high-resource parts of the year. In contrast, the Fleet Data Initiative aggregates normalized energy at the daily level before extracting a degradation rate. This gives equal weight to low- and high-irradiance times of year, which is desirable for estimating the underlying module degradation.

Another approach to estimating component degradation can be found in the statistical clear sky fitting (SCSF) approach.^[14] SCSF is an unsupervised machine learning approach which decomposes the signal of a PV power time series in various components. One of the key components is the production realized by the system in clear-sky conditions; this component does not change from year to year. Another component is a linear degradation which can be interpreted as component degradation. SCSF does not use weather or irradiance data. Similar to the PV Fleet Data Initiative methodology, this approach seeks to estimate component degradation, but there are some important differences. Like the PV Fleet initiative approach, full system outages do not influence the resulting degradation rate. However, long-term trends in partial outages and factors like tracker errors would be rejected from the analysis. Further, the approach assumes that the clear-sky resource is constant, but the clear-sky solar resource can change due to fluctuations in aerosols.^[16] This can cause bias in the degradation estimate, especially for data sets with only a few years of data.

6. Conclusion

PLR is often cited as an important metric in PV performance, and it is generally understood to encompass all loss factors. We have proposed a precise annualized definition of PLR as the change in system performance relative to the first year. The annualized definition allows for nonlinear changes to performance due to changes in degradation profile or operations and maintenance practices. It is important to note that PLR is distinct from component degradation, although component degradation is one factor that contributes to PLR. Further, PLR exclusively quantifies relative changes to energy output and does not capture initial offsets between the modeled and realized losses. When considering the losses that underly PLR, it is useful to distinguish between recoverable and nonrecoverable losses because this gives insight into which losses may be addressable through improved operations and maintenance. Quantifying PLR and its underlying losses remains an important active area of research.

Researchers and analysts should be clear about what losses their approach encompasses and the extent to which different recoverable and nonrecoverable losses contribute. We discussed three methods for evaluating loss over time in this context. The fixed-effects model^[12] assesses PLR as defined here while RdTools^[13] and SCSF^[14] both quantify DC degradation, though with some subtle differences.

The formulation of PLR we have proposed should be understood as a definition, and not a prescription for calculation. Real-world data is affected by many factors like outages and erroneous readings that must be considered when calculating PLR in practice. However, there is an important distinction between erroneous or missing data and actual losses like outages or soiling. Assessment of PLR should include actual losses, and it is important to be careful not to filter out these losses.

7. Future Work

PLR is a high-level metric and it can help in quantifying fleet-scale risk and performance. However, it is only a starting point when it comes to actionable information. More detailed analyses or measurements are needed to quantify the individual underlying loss factors. The community must continue to develop methods for quantifying individual loss factors and classifying them as recoverable or nonrecoverable.^[1] Degradation and soiling have been focuses of prior work,^[7,17] but there is still much progress to be made on decomposing the performance of real systems down to the individual loss factors in a self-consistent holistic analysis. In many cases, quantifying individual loss factors requires careful filtering. For example, quantifying module degradation requires removing the impacts of availability, maintenance, and operational downtime, clipping, snow loss, recoverable soiling, and any other factors that would mask module degradation. Developing these more detailed analyses is an important endeavor though, because quantifying the individual losses enables actionable information about operations and maintenance practices and differences between technologies.

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Conflict of Interest

The authors declare no conflict of interest.

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